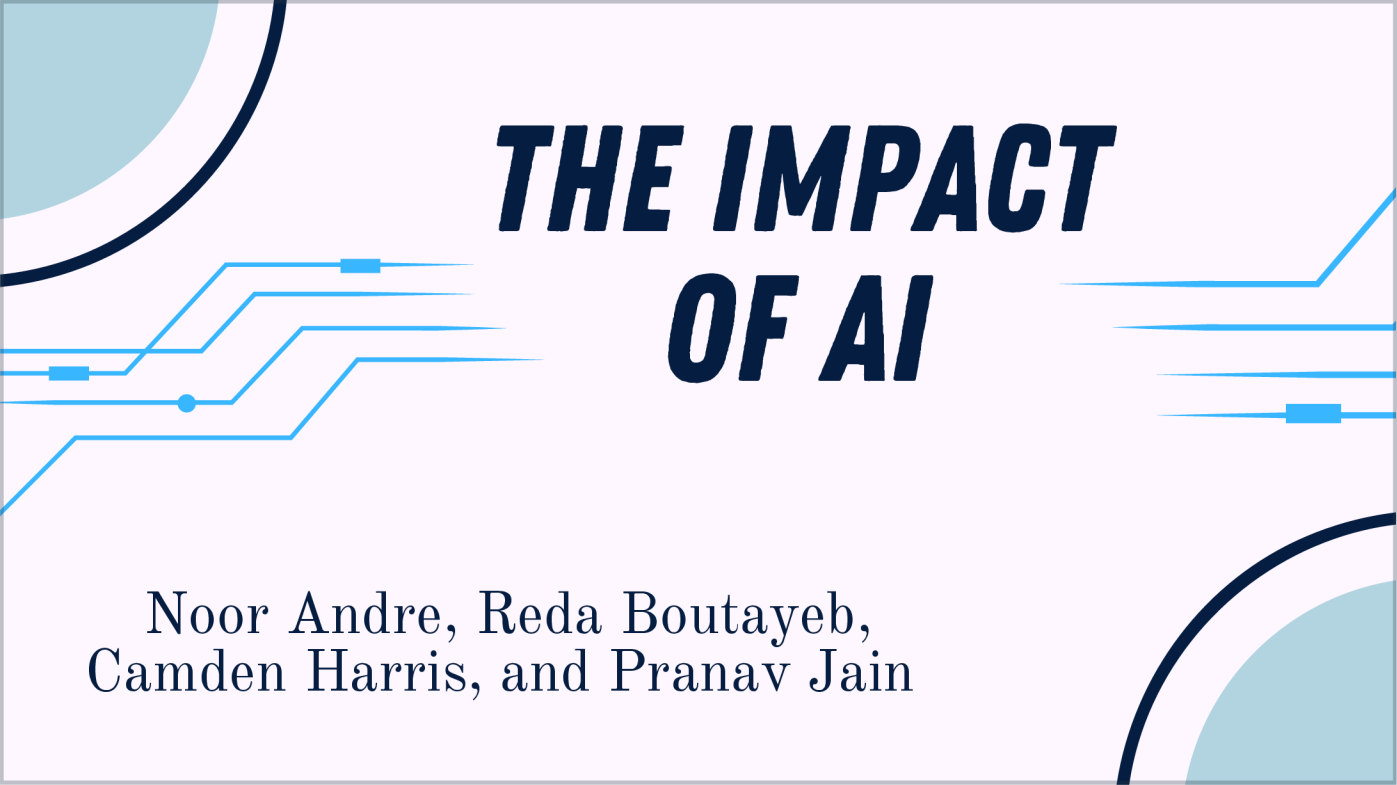


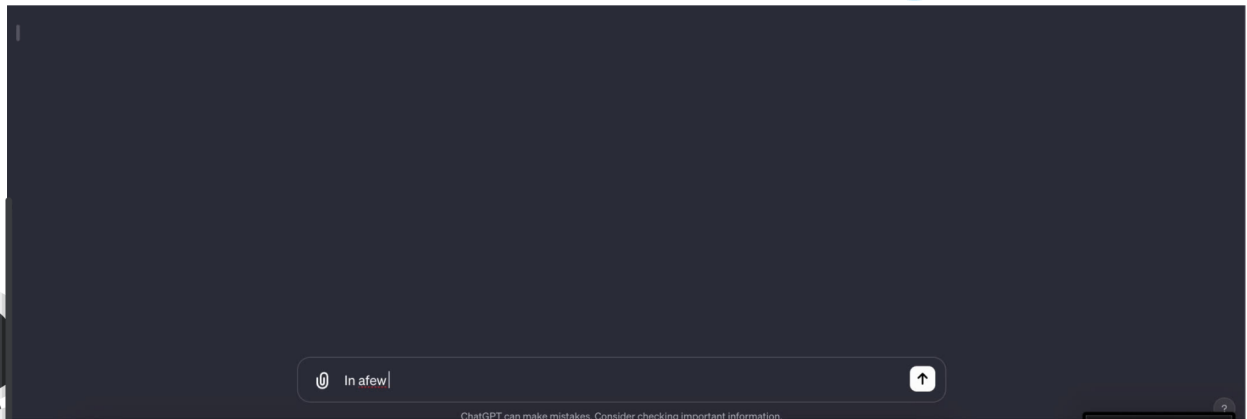


THE IMPACT OF AI

Noor Andre, Reda Boutayeb,
Camden Harris, and Pranav Jain

Hello everyone we would like to thank you all for joining us today. we are a group of students from worcester polytechnic institute in Worcester, ma. this project is a graduation requirement and we have spent the past 14 weeks researching the impact of ai on the workforce. (Then introduce ourselves.)

WILL AI TAKE OVER THE WORLD?



We thought that it would be fun to ask Artificial Intelligence , if Artificial Intelligence will take over the world?

Well, the answer we got was definitely interesting as it relates to our project.

At first, it says that AI won't take over as it lacks consciousness, however, it mentions that its impact depends on how we regulate it because its behavior depends on what it is trained to do , and that the focus should be on ethical development to ensure that it doesn't harm humans. Well, the reason that is interesting can be found in the following slides.

SPONSORS



ANSI
American National Standards Institute

- Does not develop standards
- Provides framework for fair standards development and quality conformity assessment systems



workcred
an affiliate of ANSI

- Enhances the workforce quality
- Improves the credentialing system

Our research is sponsored by the American National Standards Institute (ANSI) and its affiliate, WorkCred. For those of you who aren't familiar, ANSI is a non-profit organization established in 1918, they do not develop standards itself, but Instead, provide a comprehensive framework for fair standards development with an emphasis on maintaining their integrity.

WorkCred focuses on enhancing workforce quality by improving the credentialing system and preparing various stakeholders to effectively use it.

RISE OF AI



The term "artificial intelligence" was first introduced in 1956.



Recognized in 2023 as the "Word of the Year" by Collins English Dictionary.



Uncover the diverse implications of AI



The rapid advancement of Artificial Intelligence (AI) has become a defining feature of the modern world, reshaping industries at a remarkable pace.

Although the term “artificial intelligence” dates to 1956, ai earned the title of “word of the year” in 2023 by Collins English Dictionary, signifying the recent increased interest in AI (Stokel-Walker, 2023).

As humanity navigates this technological revolution, our research strives to uncover the diverse implications of AI on the workforce.

WHAT IS AI?



Artificial Intelligence represents a broad spectrum of technologies that utilize various computational methods to try and replicate human intelligence tasks like learning, problem-solving, and decision-making.

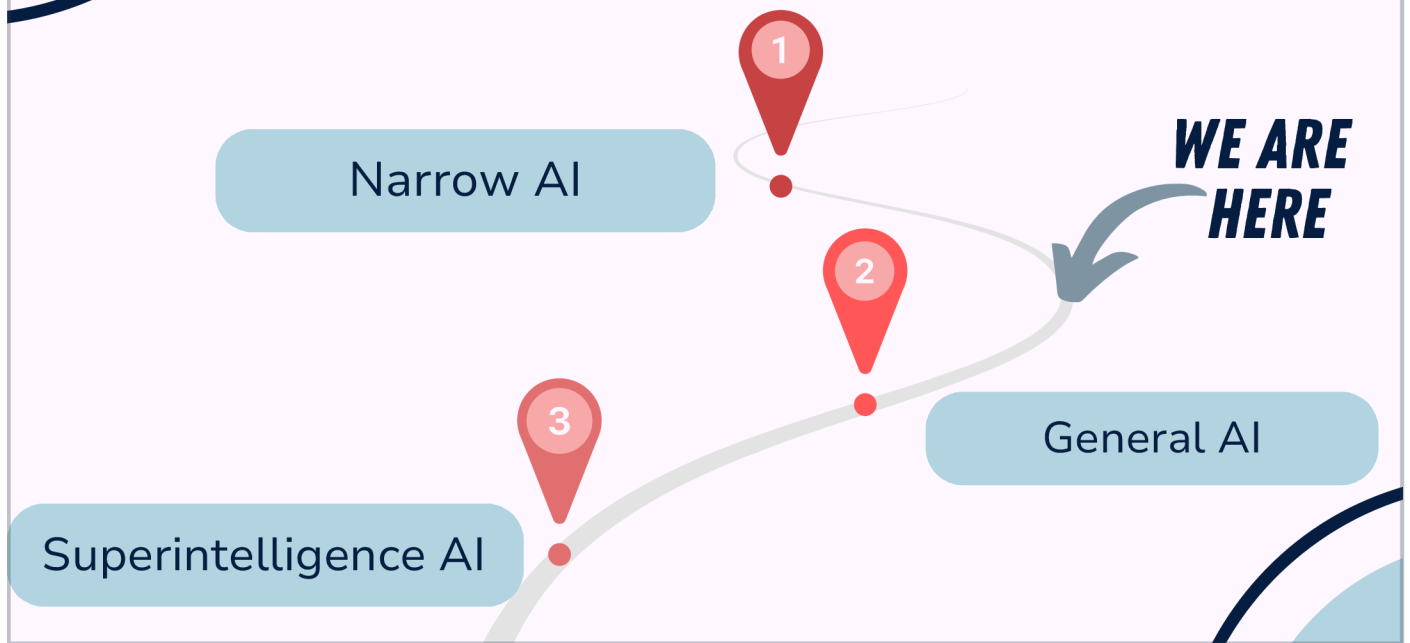


In casual terms, AI refers to machines or computer systems that are designed to perform tasks that typically require human intelligence.

ai has been around for long time and it is only getting more sophisticated, emphasizing the need for global recognition

for example at restaurants people order at kiosk instead of human interaction. ask everyone in the room if they've been to a restaurant recently and never talked to a server.

PROGRESSION OF AI

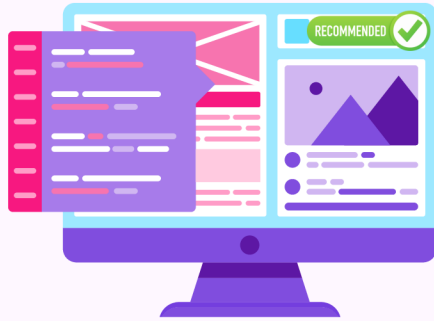




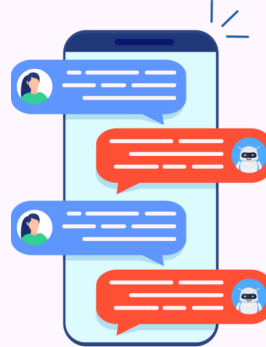
AI IN EVERYDAY LIFE



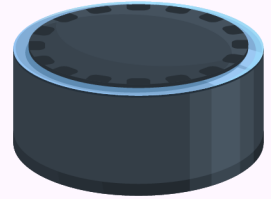
**Ordering/
purchasing
kiosk**



Website Recommendations



**Customer Service
Chatbot**



Virtual Assistant

UNRAVELING AI CONCERNS



JOB LOSS

- Historical concern began in 1930s
- Significant impact, and amplified by COVID-19
- New opportunities



MONEYWATCH >

AI eliminated nearly 4,000 jobs in May, report says

Historically, the concern about technology replacing jobs dates back to the 1930s. Recently, the COVID-19 pandemic potentially accelerated digitalization. When people transitioned to working remotely, managers recognised the increased efficiency of swapping workers with AI. This helped to speed up the implementation of AI in the workforce. Scholars who are currently debating about AI acknowledge that AI would cause potential job loss but also argue that new opportunities would be created.

ETHICAL

CONCERNS

Advances in artificial intelligence raise new ethics concerns

PBSO
NEWS
HOUR

PBS NewsHour ✓
3.85M subscribers

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“According to US Department of Justice figures, you are more likely to be arrested if you are Black than if you are white.”
(Heaven, 2020)

- Algorithmic bias
- Predictive policing and patient care
- Concerns for fairness and equity

One ethical concern is algorithmic bias. Algorithmic bias is when computer make decisions, exhibit bias or favoritism towards certain groups of people. It affects many sectors including the criminal justice and healthcare. In criminal justice, predictive policing algorithms have been found to unfairly target minorities and in healthcare, a study showed that minority patients were less likely to be referred when AI was used to prioritize patient care. This makes it crucial to address these biases for ensuring fairness and equity in representation.

CYBERSECURITY

- AI-driven cyber-attacks
- AI is a double-edged sword
- Can predict and mitigate, but also facilitate attacks



WEALTH MANAGEMENT

AI and Cybersecurity: A New Era

Sep 15, 2023

In the evolving landscape of artificial intelligence (AI), both cybersecurity teams and hackers are using AI to their advantage.



In cybersecurity, the rise of AI-driven cyber-attacks is a growing concern. While the use of AI can enhance security measures, it could also be used to identify vulnerabilities in systems at impossible speeds to humans.

2,200 cyber attacks per day
a cyber attack happening every 39 seconds

11 more people killed in crashes involving automated-tech vehicles



SCAN ME

SAFETY RISKS



- Poorly designed AI systems
- Government oversight is needed
- Human-in-the-loop concept

Poorly designed AI systems may be unsafe, especially in areas like autonomous vehicles. To address these risks, we need government oversight, industry self-regulation, and public awareness. Regular audits are necessary to examine algorithms and the 'human-in-the-loop' approach is required to ensure human oversight in important automated decision-making.

TRUSTWORTHY AI ?



- Reliability and robustness
- Europe and the U.S. collaborate on PETs
- Aim to integrate AI into society



Trustworthy AI strives to be ethical, reliable, and strong throughout its entire life cycle. Europe and the U.S. are working together on Privacy-Enhancing Technologies (PETs) for health applications, focusing on large-scale data analysis and privacy. This collaboration also explores AI's impact on climate forecasting, emergency response, medicine, and more.

LEGISLATION



"H.R. 4625 (115th): FUTURE of Artificial Intelligence Act of 2017"

- Addresses the impact of AI on society and the workforce.
- Recognizes the knowledge gap due to AI.
- Mandated the formation of an advisory group.
- The bill did not receive a vote in Congress and has not been re-introduced.



President Biden's 2023 executive order

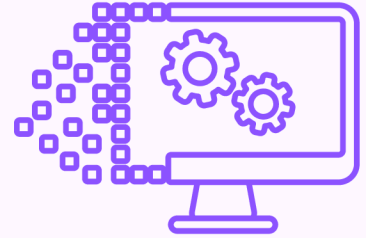
- Leads in AI development and addresses its associated risks.
- Covers AI safety and security standards, and more.

Now we are going to talk about legislation. In 2017, a bill named "Future of Artificial Intelligence" was introduced to address the significant impact of AI on the society and workforce. It recognized the knowledge gap in AI and mandated the formation of an advisory group to assess its potential impact on the workforce. This bill did not receive a vote in the next session of Congress and has not been introduced since.

This year, President Biden issued an executive order that focused on AI's development and manage its associated risks. It covered various aspects, including AI safety and security standards, privacy protection, equity and civil rights, and more.

AI'S TRANSFORMATION OF THE WORKFORCE

- Integration of AI is causing significant job role changes.
- Automation is predicted to eliminate 14% of global jobs in 15 to 20 years (Candelon et al., 2021).

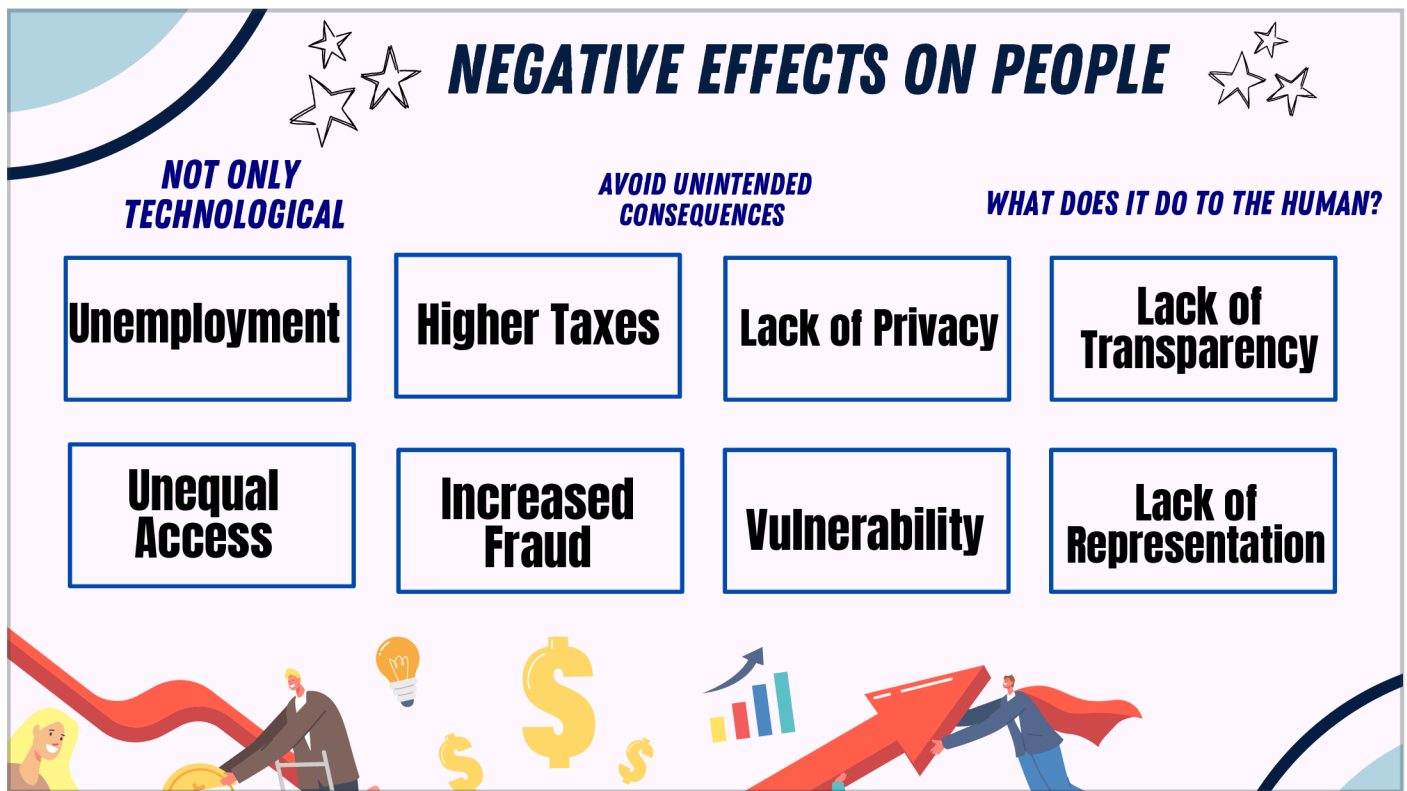


Data Analysis

Skill Gaps

Enhanced Decision-Making

Dependency and Reliability



our presentation is not solely confined to technological aspects of integrating AI, we have also considered what it will do to individual people. The concern of job loss affects unemployment and taxes. Cybersecurity affects privacy and fraud. Without ethics and legislation, certain people will be left vulnerable or under represented.

OUR GOAL

To provide a comprehensive analysis of the present-day and anticipated future impacts of Artificial Intelligence (AI) on the workforce, providing ANSI and WorkCred data-driven recommendations that will initiate their research efforts in this field.

aimed to research and provide an analysis of the impacts of AI in the workforce to provide ANSI and workcred recommendations on how to initiate their research on the topic

OBJECTIVES



Objective 1

Investigate and identify current and potential economic, social, and ethical impacts of AI on the workforce.

Objective 2

Determine implications regarding AI on the workforce from industry stakeholders' perspectives.

Objective 3

Analyze the current impacts and stakeholders' opinions to uncover potential consequences and future challenges of AI on the workforce.

METHODS

Objective 1



Google Scholar

GovInfo

Artificial
Intelligence and
Econom*

Artificial
Intelligence
and Ethic*

Artificial
Intelligence
and Social*

Artificial
Intelligence and
Accessibility

METHODS

Objective 2



Contacted 10 experts found in collaboration with our sponsors and classified them into 5 categories.

Created a set of open-ended questions tailored for each participant.

Recorded, transcribed, and summarized each of the interviews.

Policyholders/
Think Tank

Managers/ HR
Professionals

Researchers

Credentialing
Professionals

Consumer
Advocates

METHODS

Objective 3



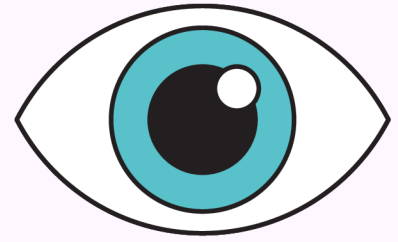
Synthesized our findings from objectives 1 and 2 to develop a comprehensive understanding of the opportunities and challenges in the evolving workforce landscape.

Identified similarities and differences between existing scholarly publications and expert perspectives on the role of AI in the workforce.

Used the synthesized information to craft recommendations.

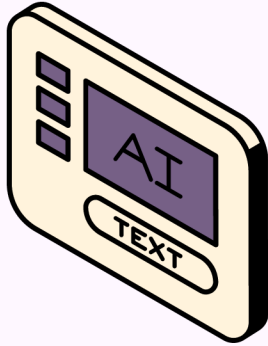


UNVEILING PARTICIPANTS PERSPECTIVES



- Varied perceptions among HR professionals and managers
- Emphasis on AI literacy and continuous learning
- Importance of human involvement for accuracy, especially in sensitive areas
- Acknowledgment of AI efficiency but concerns about job security

ADDRESSING CHALLENGES AND ETHICAL DIMENSIONS



- Rapid evolution of AI poses ethical, societal, and regulatory challenges.
- Concerns about biases in hiring and credentialing processes.
- President Biden's executive order calls for safe, secure, and trustworthy AI development.

EXPLORING SOCIETAL AND ECONOMIC SHIFTS

- AI reshaping socio-economic dynamics with potential for innovation and growth
- Risks of aggravating existing inequities
- Consensus on inclusive AI policies for equitable distribution of benefits
- Focus on marginalized communities, privacy, and data protection



PARTICIPANT VOICES



TRUST DOESN'T COME MERELY BECAUSE IT'S AN AI TOOL; WE NEED TO KNOW WHO ELSE IS USING IT AND WHO'S BEHIND IT.

A COHESIVE GOVERNANCE FRAMEWORK AT THE FEDERAL IS NECESSARY FOR AI REGULATION.

TRUSTWORTHY AI SHOULD BE A MODEL THAT DOESN'T HALLUCINATE OR POSE LEGALITY ISSUES, LIKE POTENTIAL COPYRIGHT INFRINGEMENT OR PROVIDING DANGEROUS OR INACCURATE RESPONSES.

THE NEXT INDUSTRIAL REVOLUTION

WHEN YOU THINK ABOUT ARTIFICIAL INTELLIGENCE, ETHICS ARE ALWAYS AN ISSUE, BECAUSE YOU'RE NOT DEALING WITH A CONSCIOUS ENTITY, YOU'RE DEALING WITH AN UNCONSCIOUS ENTITY, WHICH IS KIND OF A ROBOT.

COMPANIES SHOULD MAKE IT CLEAR WHEN CUSTOMERS ARE INTERACTING WITH AI.

WE REQUIRE CERTIFICATION BODIES TO PROVIDE EVIDENCE THAT THE AI PLATFORMS THEY USE ARE FAIR, AND TRANSPARENT, AND DO NOT DISADVANTAGE ANY GROUPS.

Reskilling/ AI- Literacy

WHAT SOCIETY NEEDS

- A workforce that is adaptable and technologically adept.
- A re-evaluation of job roles and skills required to perform.
- An emphasis on continuous learning and adaptability.

RECOMMENDATION 1

- Develop reskilling programs aligned with evolving AI-driven workplace demands.
 - Focus on technical skills and fostering AI literacy among the workforce.
 - Measure and plan training programs.
 - Collaborate with educational institutions, industry, and policymakers.

DESIRED OUTCOMES

- Skills improved
- Increased efficiency in the workforce
- Job security
- Boost individual's confidence in abilities



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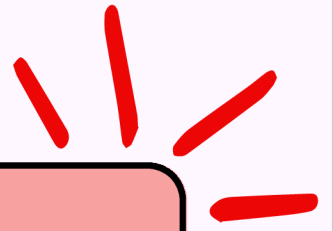
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- Job security
- Boost individual's confidence in abilities





ESSENTIAL

Ensuring that AI's integration into the workforce enhances rather than undermines human dignity and labor rights.



Regulation/Mitigation

WHAT SOCIETY NEEDS

- Custom risk assessments and ethical evaluations for each use of AI.

RECOMMENDATION 2

- Advocate specific legislation to be created to mitigate the risks.
 - Robust governance and bias mitigation to ensure responsible and transparent use.
 - Inclusive AI policies and practices that consider the diverse needs and challenges of different population groups.
 - Establish mechanisms for continuous monitoring and evaluation of AI developments.

DESIRED OUTCOMES

- Ethical use of AI
- Inclusive to all groups
- Transparency allows people to make informed decisions and foster accountability for AI benefits and limitations



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Regulation/Mitigation

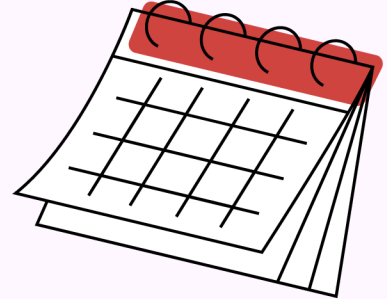
DESIRED OUTCOMES

- Ethical use of AI
- Inclusive to all groups
- Transparency allows people to make informed decisions and foster accountability for AI benefits and limitations



RESEARCH LIMITATIONS

- Companies at different stages of AI implementation.
- Confidentiality restrictions in companies.
- Variation in set of questions asked.



PERSONAL EXPERIENCE

Better understanding of AI

Learn about social aspects of the workforce

Research Skills Development

Networking & Professional Development

Lifelong memories and friends

ACKNOWLEDGEMENTS



WE WOULD LIKE TO THANK



Dr. Roy Swift, Cleo Stamatatos, Mary Saunders

Prof. Lauren Mathews and Prof. Adrienne Hall-Phillips

Interview Participants

Dean Kent Rissmiller and Worcester Polytechnic Institute

We thank our mentors Roy Swift, Cleo Stamatatos, and Mary Saunders who provided their valuable insight and expertise that greatly assisted our research. We also extend immense gratitude to our advisors, Prof. Adrienne Hall-Phillips and Prof. Lauren Mathews for their constant support and guidance. We would like to acknowledge and thank all our interview participants for sharing their opinions and experience to assist our research. We would also like to thank Associate Dean Kent Rissmiller for organizing this opportunity. Finally, to our institution Worcester Polytechnic Institute for this valuable opportunity .



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BACKUP SLIDES

NARROW OR WEAK AI

- Specific tasks
- Solving problems
- Lacks common intelligence



MACHINE LEARNING

1

SUPERVISED

2

UNSUPERVISED



Two different methods



Inspired by human thinking



Computerized learning

SUPERVISED LEARNING

This learning involves algorithms that analyze labeled data to identify patterns or make predictions (Selenko et al., 2022).

1

UNSUPERVISED LEARNING

This learning allows algorithms to explore data without any labeled responses, aiming to discover inherent structures or relationships (Selenko et al., 2022).

2

GENERATIVE AI

It refers to algorithms, like ChatGPT, capable of generating diverse content types including audio, code, images, text, simulations, and videos (Mckinsey & Company, 2023).

GPT-3 was trained on around 45 terabytes of text data—that's about one million feet of bookshelf space, or a quarter of the entire Library of Congress



<https://www.computerworld.com/article/3686616/6-surprising-facts-about-chatgpt-nobody-told-you.html>
<https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-is-generative-ai>

More than the majority of Americans know little or a lot about Generative AI.

A couple of known industries already depend on Generative AI.



GENERAL AI



- Understanding, learning, and applying knowledge
- Capable of reasoning
- Form of consciousness or emotional understanding

SUPERINTELLIGENCE AI

- Creativity and social intelligence
- Complex problem solving
- Outperforming human minds

